Kavin Manikandan

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Education

Duke University, Pratt School of Engineering **Expected Summer 2028**
Master of Engineering, Biomedical Engineering with Medical Device Design Certificate *Durham, NC*
- Certificate in Innovation & Entrepreneurship

Vellore Institute of Technology (VIT) **Aug. 2025**
*Bachelor of Technology in Biotechnology — GPA: **9.5/10** - Rank 6/284 (Top 2%)* *Vellore, India*
- **Awards & Honors:** Merit Scholarship, Top 2% of cohort (2023–24 & 2024–25); Winner, VIT Medical Hackathon
- **Extracurriculars:** Student Events Coordinator, Youth Red Cross Volunteer & SBE Community Outreach

Work Experience

Caldor Health Technologies **Aug 2025 - Aug 2026**
Quality & Regulatory Associate - Medical Devices *Coimbatore, India*
- Set up the company's first CDSCO regulatory submission pathway for a Class C device. Built a 100+ document QMS across 5 departments from scratch and got ISO 13485 certified by TÜV Nord with zero non-conformances.
- Ran a FMECA risk assessment across 11 subsystems, found 102 design risks, driving 12 resulting engineering design changes.
- Designed a compliant manufacturing setup from the ground up in 45 days, sourcing and selecting 6 vendors, specifying equipment, and defining process controls to prepare the device for production.
- Authored the core design control documentation (user needs, design inputs, and software requirements), translated clinical needs into formal engineering specifications.
- Co-wrote various funding proposals, resulting in $625,000 in funds from ICMR (Indian Council for Medical Research) under the ANRF MAHA MedTech Scheme.

TheraCUES Innovations Pvt. Ltd. **Aug 2023 - Sep 2023**
Research Intern *Bangalore, India*
- Conducted gene expression and spatial transcriptomics analysis across 30+ patient samples to study spesis biomarkers
- Co-developed a rapid diagnostic test for sepsis, reducing time-to-diagnosis by over 90%, from 24+ hours with standard culturing methods to under two hours.
- Directed the pathogen prioritization strategy for the diagnostic test, synthesizing findings from 20+ research publications to select the 10 most clinically relevant target pathogens for assay design

Research Projects

Computational Drug Discovery for a Rare Genetic Disorder (Glutaric Aciduria Type 1) **2024 - 2025**
Centre for Bio-Separation Technology, VIT *Vellore, India*
- Identified a novel therapeutic target for a rare genetic disorder and screened 10,000+ candidate compounds via computational modeling, narrowing to a shortlist of 3-5 for further MD simulation. (Publication in progress)

Genome Editing for Real-Time Tracking of Stem Cell Differentiation **2025**
Institute for Stem Cell Sciences and Regenerative Medicine *Bangalore, India*
- Engineered a CRISPR-edited stem cell line to track differentiation in real time, used it to study how environmental factors affect differentiation, improving a key developmental marker by up to 2x through protocol optimization.

Leadership

Society for Biological Engineering (SBE) - VIT **Jan 2023 - Jan 2024**
Design Head *Vellore, India*
- Led a 300-person student organization, planning 20+ professional development and networking events and hosting a podcast that reached 1,000+ listens across platforms.
- Overhauled the design department - transitioning to a Adobe + Figma ecosystem and growing team participation by 30% through a new collaborative approach, eventually driving a 150% increase in social media engagement.

Toastmasters International - VIT **Jul 2023 - Dec 2023**
Vice President, Public Relations *Vellore, India*
- Coordinated campus-wide PR campaigns, growing club membership 50% from 20 to 30 members.
- Mentored 8 junior students in public speaking, 6 of whom went on to win India-wide speech contests.

Skills & Interests

Regulatory & Quality: ISO 13485, FDA 21 CFR 820, Design Controls & Risk Management (FMECA)
Technical Tools: Minitab, Python, R, MATLAB, Fusion 360, Microsoft Office Suite (Excel, PowerPoint, Outlook) – Proficient
Life Sciences Techniques: PCR/qPCR, CRISPR/Cas9 Gene Editing, Flow Cytometry, NGS Data Analysis
Interests: Drumming & electronic music production (10,000+ online streams); Formula 1 & MotoGP enthusiast